

Education Experience:

I hold a BSc degree in Medical Physics with Physics from UCL. During this academic journey, my final project was a pivotal opportunity to develop my research skills and apply knowledge practically. The project concluded with the successful creation of a prototype that translated the recorded EMG signal from my own arm into real-time simple tones. This creative project enhanced my problem-solving skills and deepened my interest in understanding and leveraging physiological information.

I am currently pursuing an MSc in Biomedical Engineering with a specialization in the Neurotechnology stream at Imperial College. This program has provided me with structured knowledge and research skills in the field of Neuroscience. I have gained a solid foundation in basic neuroscience principles and have learned about various methodologies for recording, processing, and analyzing neural data. In terms of programming, I have utilized both MATLAB and Python for tasks such as decoding spike data and modeling neural networks, specifically in the domain of Computational Neuroscience and Brain-Machine Interface.

For my ongoing MSc project, I am focusing on the study of brain organoids that are cultured in the lab. This in vitro research aims to track the neural dynamics across early-stage organoids with the goal of future applications in studying neurodegenerative diseases. The project involves hands-on work in the wet lab, where I take part in the growth and maintenance of cerebral organoids. Moreover, I will have the opportunity to acquire neural data and gain analysis experiences. Working in a neuroscience lab would provide me with transferable skills that are applicable to this job opportunity.

Additional Laboratory Experience:

Throughout a summer internship, I had the opportunity to work in a biomedical ultrasound laboratory. This work differed from my previous experiences of being supervised on undergraduate and master's projects, as it demanded my decision-making skills and responsibility for the research advancement. Over the course of two months, I revised the initial plan and collaborated with the supplier to address any issues and gather feedback, ensuring timely completion of the assigned task.

Motivation and Future Plan:

My initial interest in the neuroscience field originated from my undergraduate project, which attempted to gather and decode information from EMG signals. This endeavor left me with lots of curiosity and many unanswered questions upon its completion. Motivated by the limitation of extracting and understanding the information from the data, I made the decision to delve into current advancements in neuroscience in order to pursue my passion.

Furthermore, I am consistently intrigued by exploring the interface and interplay between the scientific study of brain and the mechanisms underlying mental health conditions. The prospect of investigating of human brain, elucidating the connections between physiological and mental changes holds tremendous potential. I believe that this opportunity could equip me with profound expertise, enabling me to pursue a PhD and starting my own research in the future.